

Naturally Circulating Hot Water System



Steve Greenberg is a facilities energy management engineer at Lawrence Berkeley National Laboratory and a technical editor at *Home Energy*.

Q: I have been having a discussion with a fellow worker concerning our house heating systems. He claims to have a natural circulating hot-water system, in a two-story row home. He says it has no circulating pump and that temperature differential was the only reason the system worked. I did some research and all the systems I have seen on the Net show them as open systems, not closed.

Also, the pipe size seems to have a lot to do with what creates the direction of flow. That is, large pipe for risers and small ones for returns. Isn't it correct that for flow to occur, a differential pressure needs to exist? Isn't this the reason for the different pipe sizes? Please explain how this system works. I find it hard to believe that temperature alone could move water through a boiler, piping, valves, and the radiators and back to the boiler.

Pete Ianuzzi
Baltimore, Maryland

A: Hard to believe but true: Natural-convection hot-water heating systems did and most likely still do exist, though I doubt if anyone is installing new ones. They are variously known as natural-circulation, gravity circulation, and thermal-circulation systems. For such systems to work, the boiler must be at a lower elevation than any of the radiators, the supply piping must not slope down in the direction of flow, and the return piping must not slope up in the direction of flow.

The pipes and valves also need to be larger than they would be in a pumped system, since the pressure difference is typically on the order of 1 or 2 inches of water column (versus many feet of water in a pumped system).

But that little amount of differential pressure, created solely by the density difference between hot and cold water (hot water rises, cold water sinks) is enough to make it work. The pressure difference is also dependent on the height between the boiler and the radiator, so the upper-floor radiators will (all other things being equal) get more flow than the lower-floor ones.

These systems can work either open or closed, but if they are closed, there would need to be an expansion tank, pressure relief valve (if the vent of an open system couldn't be valved off from the boiler), and so on. I think the

reason they were typically built open is consistent with the overall simplicity of the system.

There's no need for the pipes to be different sizes. You are correct; differential pressure causes the flow, but the differential pressure available is due solely to the temperature and elevation differences between the boiler to the radiator. How much flow depends on that differential pressure and on the restriction presented by the piping, radiators, and valves. Smaller pipe will result in lower flow, other factors being equal. The warmer water is a bit less dense, but that difference—for 180°F versus 140°F, it's a difference of only about 1%—wouldn't justify different pipe sizes.

There are other examples of gravity circulation water systems. These include systems for domestic hot water circulation (still in use in older, typically residential, multistory buildings) and solar hot water (the storage tank must be above the collector). The latter are known as a thermosiphon systems.

For more information:

To learn more about thermosiphon systems go to www.eere.energy.gov/de/solar_hotwater.html.) Several companies manufacture these systems. See, for example, www.solahart.com.